WORKED SOLUTION

STEP I 2003, Question 10

Mechanics • Centre of Mass of a Lamina

STEP I 2003, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s1-q10

The key idea

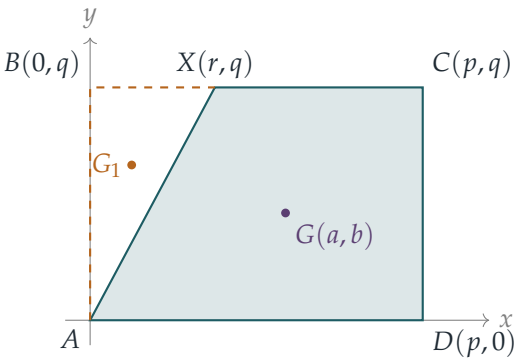
A shape with a piece removed is best handled by *subtraction*: the rectangle's moment about each axis equals the trapezium's moment plus the triangle's moment, with masses proportional to areas. For the suspension, a lamina hanging freely from a point comes to rest with its centre of gravity directly below that point, so the line AG is vertical. The 45° condition then turns into a very simple relation between a and b.

Solution

We model the lamina as uniform and thin, so the mass of any piece is proportional to its area and the centre of gravity of a triangle is at its centroid (the mean of its vertices).

The centre of gravity

With the given axes, $A = (0,0)$, $D = (p,0)$, $C = (p,q)$, and so $B = (0,q)$. Since X lies on BC with $BX = r$, we have $X = (r,q)$, where $0 \leq r \leq p$.



The three pieces are as follows.

- The rectangle $ABCD$: area pq, centre of gravity $(\frac{p}{2}, \frac{q}{2})$.
- The triangle ABX (removed): area $\frac{1}{2}qr$, centre of gravity G_1 at the mean of $(0,0)$, $(0,q)$ and (r,q), which is $(\frac{r}{3}, \frac{2q}{3})$.
- The trapezium $AXCD$ that remains: area $pq - \frac{1}{2}qr = \frac{1}{2}q(2p-r)$, centre of gravity $G = (a,b)$.

The rectangle's moment about each axis is the sum of the moments of its two parts. Taking moments about the y-axis (that is, comparing x-coordinates),

$$pq \cdot \frac{p}{2} = \frac{qr}{2} \cdot \frac{r}{3} + \frac{q(2p-r)}{2}\, a.$$

Dividing by $\frac{q}{2}$ gives $p^2 = \frac{r^2}{3} + (2p-r)a$, and since $2p - r \geq p > 0$ we may divide by it:

$$a = \frac{3p^2 - r^2}{3(2p-r)}.$$

Similarly, about the x-axis,

$$pq \cdot \frac{q}{2} = \frac{qr}{2} \cdot \frac{2q}{3} + \frac{q(2p-r)}{2}\, b,$$

so $pq = \frac{2qr}{3} + (2p-r)b$ and

$$b = \frac{q(3p-2r)}{3(2p-r)}.$$

As a quick check, $r = 0$ (nothing cut off) gives $a = \frac{p}{2}$, $b = \frac{q}{2}$, the centre of the rectangle.

Suspending from A

When the lamina hangs freely from A and is in equilibrium, only two forces act on it: its weight, through G, and the reaction at A. Two forces in equilibrium must act along the same line, so G lies vertically below A.

Ask yourself How does "AD is at 45° below the horizontal" become an equation in a and b?

Since AG is vertical and AD makes 45° with the horizontal, AD also makes 45° with AG; that is, the angle GAD is 45°. In our axes AD is the positive x-axis and $G = (a,b)$ with $a, b > 0$ (it lies inside the trapezium), so $\tan \angle GAD = b/a$. Hence $b/a = 1$, that is,

$$a = b.$$

Using the expressions above and cancelling the common positive denominator $3(2p-r)$,

$$\begin{aligned} & 3p^2 - r^2 = q(3p-2r) \\ \iff \quad & r^2 - 2qr + 3pq - 3p^2 = 0. \end{aligned}$$

By the quadratic formula,

$$r = q \pm \sqrt{q^2 - (3pq - 3p^2)} = q \pm \sqrt{q^2 - 3pq + 3p^2}.$$

The quantity under the root is $(q - \frac{3}{2}p)^2 + \frac{3}{4}p^2 > 0$, so both roots are real and distinct. Write $s = \sqrt{q^2 - 3pq + 3p^2} > 0$.

Choosing the sign. Since X lies on the side BC, which has length p, we need $0 \le r \le p$. We show that the root $q + s$ is always bigger than p.

- If $q \ge p$, then $q + s > q \ge p$, because $s > 0$.
- If $q < p$, then $p - q > 0$ and

$$s^2 - (p - q)^2 = (q^2 - 3pq + 3p^2) - (p^2 - 2pq + q^2) = 2p^2 - pq = p(2p - q) > 0,$$

since $q < p < 2p$. So $s > p - q$ and hence $q + s > p$.

Either way the plus sign would put X beyond C, off the lamina, so it is impossible. Therefore

$$r = q - \sqrt{q^2 - 3pq + 3p^2},$$

as required.

For completeness: this root really does lie in $[0, p]$ exactly when $p \le q \le 2p$. Indeed $q - s \ge 0 \iff q^2 \ge s^2 \iff 3p(q - p) \ge 0 \iff q \ge p$, and (when $q > p$) $q - s \le p \iff (q - p)^2 \le s^2 \iff p(2p - q) \ge 0 \iff q \le 2p$. So the situation described can only arise for such shapes, which matches intuition: the uncut square ($q = p$, $r = 0$) hangs from a corner with its diagonal vertical, and at the other extreme ($q = 2p$, $r = p$) the shape is the right-angled triangle ACD, whose centroid $(\frac{2p}{3}, \frac{q}{3})$ again has $a = b$.

Checks

- Setting $r = 0$ recovers the rectangle's centre $(\frac{p}{2}, \frac{q}{2})$, and $r = p$ gives $(\frac{2p}{3}, \frac{q}{3})$, the centroid of triangle ACD.
- With $p = 1$, $q = 1.5$ the formula gives $r = 1.5 - \sqrt{0.75} \approx 0.634$, and then $a = b \approx 0.634$, so AG is indeed at 45° to AD. The other root, ≈ 2.37, exceeds $p = 1$.
- We also checked the centre-of-gravity formulae against the general polygon centroid formula for thousands of random p, q, r.

Common slips

- Using the triangle's centroid as $(\frac{r}{3}, \frac{q}{3})$: two of its vertices are at height q, so its centroid is at height $\frac{2q}{3}$.

- Adding the triangle's moment instead of subtracting it: the trapezium is the rectangle *minus* the triangle.
- Rejecting the plus sign with a vague "r must be small". The real reason is $r \leq p$, and that needs a short argument, split by whether $q \geq p$.

Beyond STEP

Centres of mass by subtraction are a discrete version of computing $\iint x \, \mathrm{d}A \big/ \iint \mathrm{d}A$, and at university you meet the same idea with double integrals, for laminas of varying density and for solids. The hanging condition is the simplest case of a general principle: a rigid body pivoted at a point is in stable equilibrium when its centre of mass is directly below the pivot, where its potential energy is least. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats centres of mass and rigid-body equilibrium carefully.

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